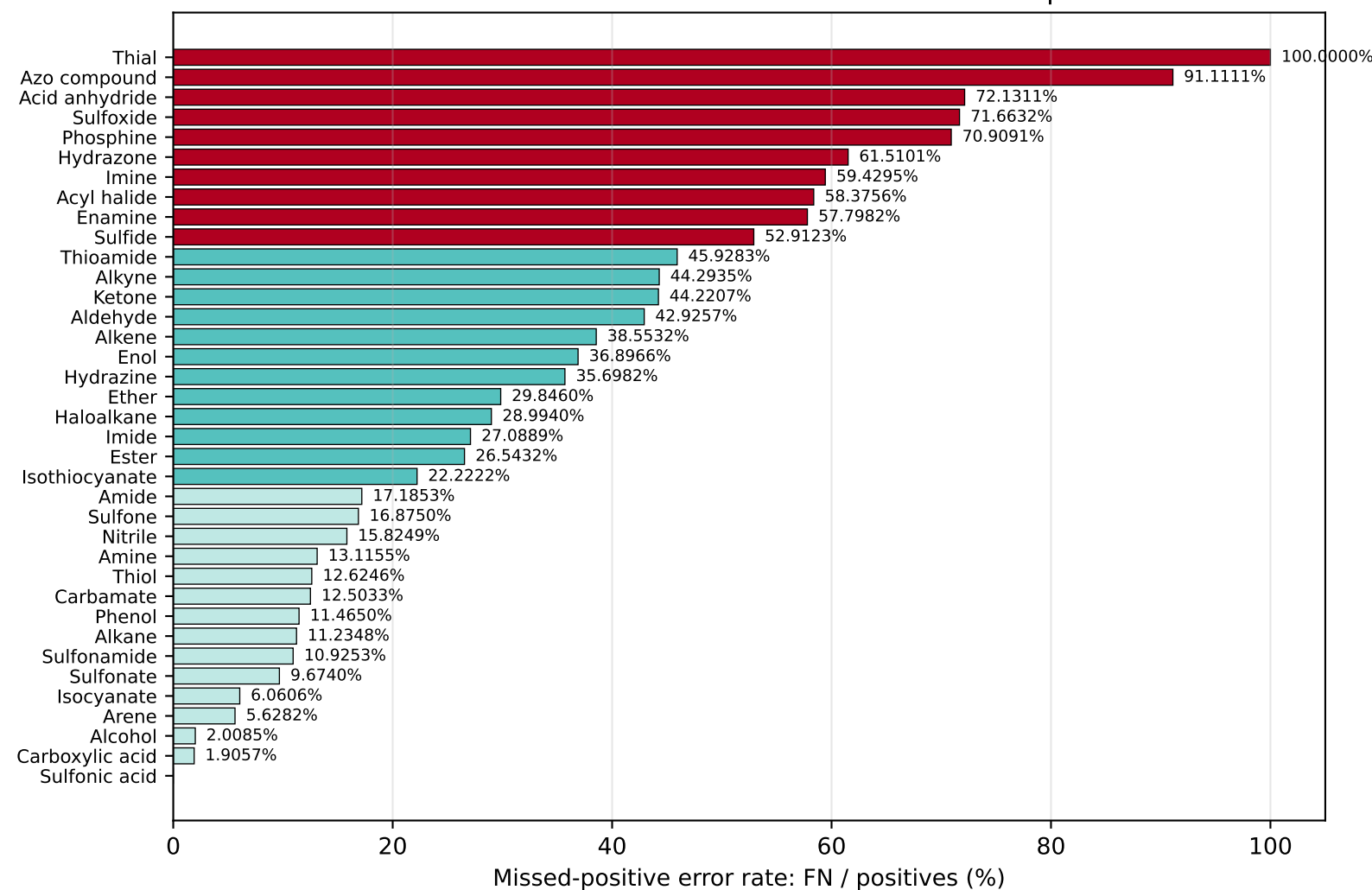
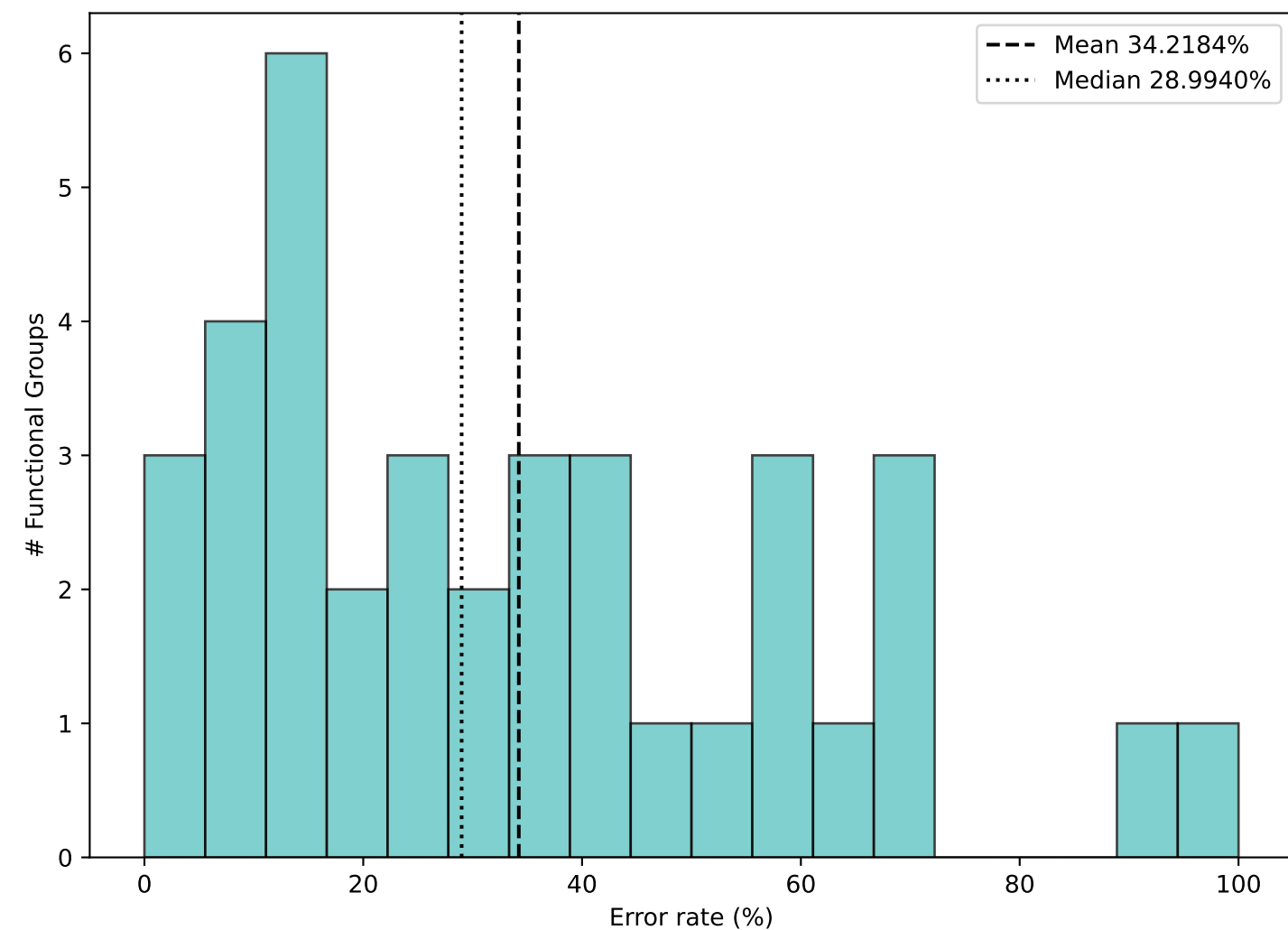


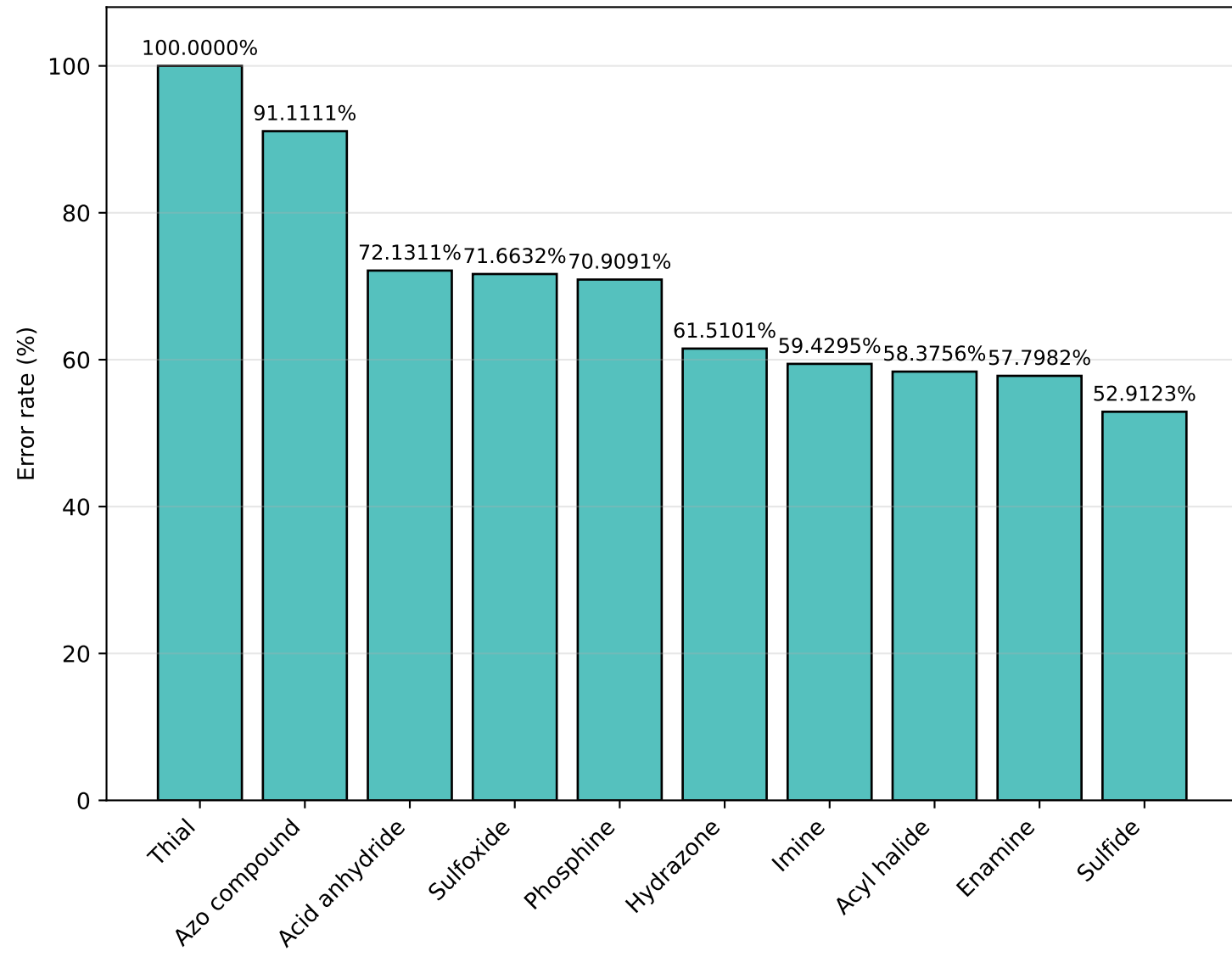
MPS IR Validation Set - All 37 Functional Groups



Error-Rate Distribution



Top 10 Problem Groups



MPS IR VALIDATION SET ERROR ANALYSIS

Global metrics:

F1 micro:	0.8429
F1 macro:	0.6621
Precision micro:	0.8685
Recall micro:	0.8188
Hamming accuracy:	0.9606

Error-rate statistics:

Mean:	34.2184%
Median:	28.9940%
Std:	26.2086%
Max:	100.0000% (Thial)
Min:	0.0000% (Sulfonic acid)

Problem classes:

Severe (>50%):	10 groups
High (20-50%):	12 groups
Moderate (5-20%):	12 groups
Low (<=5%):	3 groups